

- 6 (a) A student investigated the effect of changing the surface area to volume ratio on diffusion.
- Two different-sized blocks of agar, **X** and **Y**, were made.
 - The agar contained Universal Indicator solution.
 - Universal Indicator solution changes colour when acid is added.
 - The blocks were placed in dilute hydrochloric acid at the same temperature.
 - The student timed how long it took for each block to change colour completely.

Blocks **X** and **Y** are shown in Fig. 6.1. All dimensions are in cm.

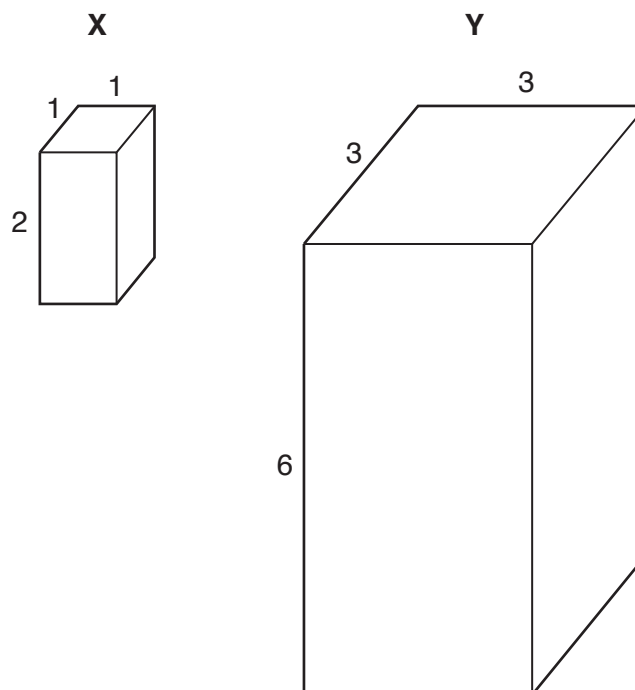


Fig. 6.1

- (i) The surface area to volume ratio of block **X** is 5:1.
- Calculate the surface area to volume ratio of block **Y**.
- Show your working.

- (ii) The student observed that block **X** changed colour completely in a much shorter time than block **Y**.

Explain why.

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- (iii) Suggest how the results of this investigation help to explain why plants need a transport system.

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- (b) An experiment was carried out by the student to investigate the ability of reducing sugars to diffuse through Visking tubing. Fig. 6.2 shows the apparatus used.

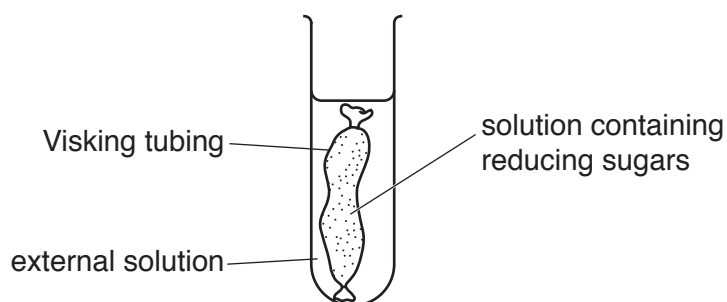


Fig. 6.2

At the start of the experiment, the external solution did not contain any reducing sugars.

At intervals, the student tested for the presence of reducing sugars, both within the Visking tubing and in the external solution.

Name the reagent that is used to test for the presence of reducing sugars.

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[Total: 7]